Prasidhi Nandwani

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Education

Integrated M.Tech in Artificial Intelligence, Vellore Institute of Technology 2022 – 2027

- CGPA: 8.73/10

Technical Skills

- **ML & Data:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SQL, EDA, ETL pipelines, ONNX, model optimization
- **NLP & Unstructured Data:** librosa, MFCCs, OpenCV, nibabel, audio/image/medical imaging pipelines
- **Deployment & Cloud:** FastAPI, Flask, REST APIs, Docker, AWS, Azure, Linux, Git, Kubernetes
- **Languages:** Python, SQL, Java, JavaScript, Bash

Projects

- **Autism Screening Using Infant Cry Sounds** [Live Demo | GitHub] *Jan 2026*
  Python, librosa, TensorFlow / PyTorch, scikit-learn, Flask, Docker
  – Built an end-to-end ML pipeline on unstructured audio data — including feature extraction (MFCCs), data preprocessing, model training, evaluation, and iterative experimentation across multiple classifiers.
  – Deployed a REST API with Flask and Docker for real-time inference, making predictions accessible over the web.
- **Swerve Detection for Driver Assistance Systems** [GitHub] *Nov 2025*
  Python, PyTorch / TensorFlow, scikit-learn, ONNX, Quantization
  – Processed streaming time-series sensor data and ran comparative experiments across 1D-CNN, LSTM, and Isolation Forest models to identify the best-performing approach for anomaly detection.
  – Optimized latency via ONNX export and quantization; achieved near real-time inference throughput.
  – Merit Distinction — Edge Impulse Hackathon.
- **Brain Tumor Segmentation Using Instant-NGP Hash Grid Encoding** [GitHub] [Blog] *May 2026*
  Python, PyTorch, nibabel, Plotly, Google Colab, BraTS 2020
  – Adapted NVIDIA's Instant-NGP encoding from 3D scene reconstruction to volumetric MRI segmentation; designed subject-level train/val split and evaluated using Dice scores across three tumour sub-regions.
  – Achieved Dice scores of 0.704 / 0.654 / 0.769 (WT/TC/ET) on BraTS 2020 with a 2.1M-parameter model trained on a single T4 GPU across 10 subjects.
- **Jyoti: Currency Classification with Hindi Voice Output** [GitHub] *Aug 2025*
  Python, OpenCV, TensorFlow / Keras, TFLite, Hindi TTS
  – Curated and augmented image datasets to improve classifier robustness; trained and evaluated a lightweight CNN for real-time currency recognition with Hindi audio feedback.
  – Awarded **Dr. Kalam Young Achiever Award** for accessibility-focused AI deployment.

Research Publications

- **Lightweight CNN with SE Attention for Brain Tumor Classification** (2025, under review). Compact SE-CNN optimized for deployment; 99.24% accuracy.
- **Comparative Study on Output Modes in Object Detection & Distance Estimation**, IATMSI 2025. DOI
- **Speech-Based Visual Aid for the Blind Eye using Deep CNN**, ETNCC 2024. DOI

Achievements

- Dr. Kalam Young Achiever Award, 2025 — *Jyoti: Currency Classification with Hindi Voice Output*
- Raman Research Award, VIT Bhopal, 2025 — Outstanding undergraduate research in ML
- Merit Distinction — Edge Impulse Hackathon (Edge-Based ADAS System)
- UIDAI Data Hackathon 2026 — Finalist

Certificates

- Applied Machine Learning in Python (Coursera)
- Simulink Onramp Course (MathWorks)
- Cloud Computing (Elite + Silver) (NPTEL)